Network Digital Twin for Real-Time Performance Diagnosis and Congestion Mitigation

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Problem Statement

Modern communication networks are experiencing unprecedented growth in traffic volume, driven by cloud computing, video streaming, IoT deployments, and real-time applications. Traditional network management approaches rely on reactive monitoring and manual intervention, which are insufficient to address the dynamic nature of contemporary network congestion and performance degradation. Existing systems lack the capability to predict and mitigate congestion proactively before it impacts end users, resulting in service disruptions, increased latency, packet loss, and degraded Quality of Service. A Network Digital Twin offers a transformative solution by creating a high-fidelity virtual replica of the physical network that synchronizes in real time with live telemetry data. By integrating machine learning-based predictive analytics within this digital twin framework, it becomes possible to detect emerging congestion patterns early, simulate mitigation strategies safely, and deploy automated corrective actions such as traffic rerouting, load balancing, and dynamic bandwidth adjustment. This project aims to develop a real-time Network Digital Twin capable of predicting congestion and mitigating performance degradation using machine learning and network telemetry.

Objectives

The objectives of the proposed project are:

1. To design and implement a Network Digital Twin using Mininet that replicates the behavior of a communication network.
2. To collect real-time network telemetry such as bandwidth utilization, latency, throughput, and packet loss using iperf and network monitoring tools.
3. To develop a machine learning model using historical network datasets such as the GEANT dataset to predict congestion and network performance degradation.

4. To integrate real-time telemetry with the digital twin for continuous network performance diagnosis.
5. To implement congestion mitigation techniques such as rerouting, load balancing, and bandwidth adjustment within the digital twin.
6. To evaluate network performance before and after mitigation using key performance metrics.
7. To demonstrate a real-time predictive congestion mitigation framework using Network Digital Twin technology.

Literature survey

Year	Name	Focus of Paper	Key Point	Method Used	Data/Parameter
2025	APDT: A Digital Twin for Assessing Access Point Characteristics in a Network	Focuses on client density and traffic-congestion related issues.	Development of APDT which can predict short-term traffic surges, leading to improved QoS (Quality of Service) and reduced traffic congestion.	1. Ruckus SmartZone controller's RESTful API is used for network monitoring purpose 2. Postman was employed to explore API endpoints, test latency, and verify data formats. 3. NS-3 simulator employed to assess whether real-world network dynamics was effectively replicated in a digital environment	Operational data such as signal strength, client density, reception rate, transmission rate, airtime utilization, noise level, client associations, throughput, and connection quality is collected in real time through the Ruckus SmartZone Controller API
2024	**Real**-time update algorithms for digital twin models of distribution network equipment under internet of things and optical imaging technology	real-time updating of digital twin models for distribution network equipment using IoT and optical imaging data, whereas the given topic focuses on leveraging a Network Digital Twin for real-time performance diagnosis and congestion mitigation in communication networks.	emphasizes accurate synchronization of the digital twin with physical assets for timely monitoring and analysis, while the given topic emphasizes diagnosing network performance issues and proactively mitigating congestion.	Real-time update algorithms that integrate IoT sensor readings and optical imaging to maintain an accurate digital twin, whereas the given topic uses real-time telemetry, network modeling, and feedback loops within a digital twin for congestion detection and control.	uses sensor data, imaging measurements, and equipment operational metrics to update the twin, while the given topic typically uses network metrics such as delay, jitter, throughput, queue lengths, and traffic load.
2021	When Digital Twin Meets Network Softwarization in the Industrial IoT: Real-Time Requirements Case Study	The Industrial IoT paper focuses on integrating Network Digital Twin with SDN to satisfy strict real-time communication requirements in industrial environments, whereas the congestion-mitigation topic focuses on using a Network Digital Twin for real-time performance monitoring and dynamic congestion control in communication networks.	The study emphasizes protocol-level validation and deterministic timing guarantees, while the congestion-mitigation approach emphasizes adaptive diagnosis and proactive traffic management based on live network conditions.	employs SDN-based network softwarization combined with simulation and protocol evaluation (e.g., MAC scheduling analysis), whereas the congestion-mitigation approach uses real-time telemetry collection, network modeling, and predictive analytics or machine learning for performance diagnosis.	The paper evaluates latency, packet delivery ratio, and freshness indicators under different MAC protocols, whereas the congestion-mitigation topic analyzes throughput, delay, jitter, queue length, bandwidth utilization, and packet loss metrics.

2023	A Review on Congestion Mitigation Techniques in Ultra-Dense Wireless Sensor Networks: State-of-the-Art Future Emerging Artificial Intelligence-Based Solutions	focuses on surveying traditional and AI-based congestion mitigation techniques in ultra-dense wireless sensor networks, whereas the Network Digital Twin topic focuses on leveraging a virtual network replica for real-time performance diagnosis and congestion mitigation.	classification and evaluation of congestion control strategies including emerging artificial intelligence approaches, while the Network Digital Twin approach emphasizes real-time monitoring, prediction, and proactive congestion management using synchronized virtual–physical network models.	systematic literature analysis and comparative evaluation of existing congestion control algorithms, whereas the Network Digital Twin approach uses real-time telemetry integration, network modeling, and possibly machine learning within a digital twin framework.	packet loss, throughput, energy consumption, buffer occupancy, latency, and network lifetime, whereas the Network Digital Twin topic typically relies on real-time parameters like delay, jitter, queue length, bandwidth utilization, and traffic load for predictive diagnosis.
2022	Digital Twin for Networking: A Data-Driven Performance Modeling Perspective	data-driven Digital Twin framework for networking that leverages machine learning–based performance modeling to emulate and predict network behavior, whereas the given topic emphasizes real-time Network Digital Twin implementation for immediate performance diagnosis and congestion mitigation.	Predictive performance modeling, traffic matrix estimation, model calibration using historical telemetry, and virtual network emulation, in contrast to the given topic which prioritizes real-time KPI monitoring, congestion detection, adaptive routing, and closed-loop SDN/NFV-based control.	statistical regression, supervised learning, deep neural networks, and data-driven traffic analysis for constructing performance models, while the given topic integrates real-time state estimation, anomaly detection algorithms, and feedback-driven congestion control mechanisms.	historical datasets including throughput statistics, delay distributions, packet loss ratios, traffic matrices, and topology configurations, whereas the given topic depends on real-time telemetry parameters such as latency, jitter, queue length, bandwidth utilization, and link congestion indicators.
2023	A digital twin based framework for detection, diagnosis, and improvement of throughput bottlenecks	Digital Twin framework to detect, diagnose, and improve throughput bottlenecks in networks, while the given topic focuses on leveraging a Network Digital Twin for real-time performance diagnosis and congestion mitigation across network traffic.	Identifying and resolving throughput constraints using a virtual network replica, whereas the given topic emphasizes real-time monitoring and adaptive congestion control for overall network performance.	Digital Twin model of the network combined with simulation and analysis to detect bottlenecks and evaluate improvement strategies, while the given topic uses real-time telemetry, network modeling, and feedback loops within a digital twin for predictive congestion management.	throughput, packet loss, latency, and link utilization within the twin, whereas the given topic typically considers delay, jitter, queue lengths, traffic load, and bandwidth utilization for real-time congestion diagnosis.
2025	A Differentiable Digital Twin of Distributed Link Scheduling for Contention-Aware Networking	Built an analytical network digital twin (NDT) that predicts link duty cycles from network context.	Digital Twin predicts congestion and link duty cycles efficiently using analytical modeling.	Analytical Digital Twin with conflict graph modeling and gradient-based optimization techniques.	Network topology, link quality, traffic load, link capacity, and contention probability.
2025	Towards a Robust Transport Network With Self-adaptive Network Digital Twin	Self-adaptive implementation of a novel NDT architecture designed to provide accurate delay predictions, even under fluctuating traffic conditions. This architecture addresses an essential challenge, underexplored in the literature: improving the resilience of data-driven NDT platforms against traffic variability and improving synchronization between the	Uses concept drift detection to maintain synchronization between physical network and digital twin.	Graph Neural Network-based Digital Twin with telemetry monitoring and retraining triggered by concept drift detection.	Telemetry data including delay, jitter, traffic patterns, and network state collected from simulated network environments.

		VTwin and its physical counterpart.			
2024	GLANCE: Graph-based Learnable Digital Twin for Communication Networks	Develop a learnable Digital Twin using Graph Neural Networks (GNN) to predict network performance efficiently.	A learnable digital twin predicts network KPIs much faster than traditional simulation.	Graph Neural Networks (GNN), multi-task learning, transfer learning trained using NS-3 simulation data.	Network topology, traffic flow patterns, and performance KPIs such as throughput and latency.
2025	A Deep Reinforcement Learning-Based TCP Congestion Control Algorithm: Design, Simulation, and Evaluation	Develop a Deep Reinforcement Learning (DRL)-based TCP congestion control algorithm to optimize network performance.	Uses Deep Q-Network (DQN) to dynamically adjust congestion window based on network state.	Deep Reinforcement Learning (DQN) integrated with TCP congestion control, implemented in NS-3 simulator.	Network telemetry such as RTT, packet loss, congestion window size, throughput collected from NS-3 simulation.
2025	Exploiting Age of Information in Network Digital Twins for AI-Driven Real-Time Link Blockage Detection	Enhances real-time network performance diagnosis in wireless networks by maintaining synchronization accuracy between physical network and digital twin using Age of Information (AoI). The work focuses on early detection of link blockage conditions to prevent performance degradation and congestion in dynamic environments such as 5G/6G networks.	Introduces AoI-aware Digital Twin learning framework where freshness of collected network data is prioritized during AI model training, enabling accurate and timely detection of link blockage before communication failure occurs. Improves reliability of real-time twin decision making.	Digital Twin environment modeling combined with Deep Learning-based blockage prediction. Ray-tracing simulation generates virtual wireless data, followed by AoI-weighted neural network training for real-time link state classification (LoS/NLoS detection).	Channel State Information (CSI), Signal-to-Noise Ratio (SNR), propagation delay, packet transmission rate, environmental obstruction data, user mobility information, timestamp updates, and Age of Information metric representing data freshness.
2025	Differentiable Digital Twin of Distributed Link Scheduling for Contention-Aware Networking	This paper focuses on developing a differentiable Network Digital Twin capable of modeling distributed wireless link scheduling in contention-aware networking environments. The proposed digital twin accurately represents interactions among competing wireless links and enables real-time performance diagnosis by predicting congestion caused due to interference and shared spectrum usage in multi-hop networks.	The work introduces a differentiable digital twin framework that replaces traditional packet-level simulation with a learning-based analytical model capable of estimating network performance efficiently. The digital twin captures link interference, contention behavior, and scheduling dependencies, enabling faster prediction of throughput degradation and congestion scenarios while supporting intelligent and distributed network control decisions.	The authors construct a mathematical digital twin using conflict graph modeling to represent interference relationships among network links. A distributed link scheduling mechanism based on a weighted Luby's algorithm is integrated with differentiable optimization techniques, allowing gradient-based learning and iterative refinement of link activation probabilities for efficient contention resolution.	The model utilizes network topology information, interference relationships between links, link duty cycles, channel contention probability, transmission scheduling decisions, throughput estimation metrics, and spectrum sharing characteristics to accurately mirror real network behavior within the digital twin environment.

2024	TwiNet: Connecting Real World Networks to their Digital Twins Through a Live Bidirectional Link	This paper proposes TwiNet, a real-time Network Digital Twin framework that establishes a live bidirectional connection between physical wireless networks and their virtual digital twin environments. The work focuses on enabling continuous synchronization for real-time performance monitoring, diagnosis, and adaptive optimization of wireless network behavior under changing traffic and interference conditions.	The proposed system enables real-time mirroring of network traffic and spectrum conditions between the physical network and its digital twin using low-latency communication mechanisms. The digital twin continuously analyzes network states, detects abnormal configurations, and supports rapid deployment of optimized control strategies to improve network performance and reduce congestion effects.	The authors design a bidirectional synchronization framework using MQTT-based communication to transfer live network data between physical and virtual environments. Machine learning models are trained within the digital twin using synthetic and real-time data, allowing adaptive optimization and rapid reconfiguration of network parameters when congestion or interference is detected.	The system utilizes real-time wireless traffic measurements, spectrum utilization data, user equipment behavior, signal interference levels, latency metrics, packet transmission statistics, and network configuration parameters collected from operational wireless environments.
2025	Network Traffic Prediction for Multiple Providers in Digital Twin-Assisted NFV-Enabled Network	This paper proposes a Digital Twin-assisted Network Function Virtualization (NFV) architecture that enables real-time performance analysis and traffic prediction across multi-provider communication networks. The digital twin continuously mirrors network behavior and predicts traffic variations to prevent congestion and improve dynamic resource allocation in virtualized network infrastructures.	The proposed framework utilizes a synchronized Network Digital Twin to analyze traffic demand variations generated by different service providers and service function chains. By forecasting future traffic load conditions, the system enables proactive congestion mitigation through intelligent resource orchestration and adaptive bandwidth allocation before performance degradation occurs.	The authors implement a Digital Twin-assisted NFV environment combined with machine learning-based traffic prediction models. Real network operational data are replicated within the virtual twin, allowing predictive analysis and optimization of network resource management decisions in real time.	Network traffic flow statistics, service function chain requests, bandwidth utilization levels, packet transmission rate, virtual network resource allocation data, latency measurements, and provider-specific traffic variation patterns are used for performance modeling and congestion prediction.
2022	Building a Digital Twin for Network Optimization Using Graph Neural Networks	This paper proposes a Network Digital Twin model called TwinNet that enables real-time performance analysis and optimization of communication networks by accurately modeling relationships between network topology, routing configuration, traffic load, and queue scheduling policies. The digital twin predicts network performance metrics such as delay and congestion conditions without requiring real network experimentation.	The proposed Digital Twin leverages Graph Neural Networks to learn complex dependencies among network nodes and traffic flows, allowing accurate estimation of Quality of Service parameters. The virtual twin performs what-if analysis and predicts congestion scenarios, enabling proactive optimization of routing and scheduling decisions to maintain network performance stability.	The authors design a GNN-based Digital Twin framework trained using synthetic and real network traffic traces. The model learns topology-aware representations and performs performance estimation through deep learning-based inference instead of traditional simulation approaches.	Network topology structure, routing configuration data, traffic matrices, queue scheduling policies, packet delay measurements, jitter values, bandwidth utilization, and Quality of Service performance indicators are used for training and validating the digital twin model.

2026	Predicting Networks Before They Happen: Experimentation on a Real-Time V2X Digital Twin	The paper focuses on developing a real-time Network Digital Twin framework capable of predicting network behaviour before congestion or performance degradation occurs in dynamic communication environments such as V2X networks.	Introduces a predictive Network Digital Twin that synchronizes real-world network data with a virtual twin to forecast signal quality, link blockage, and performance variations in advance rather than reacting after congestion occurs.	Real-time Digital Twin modeling integrated with mobility tracking, deterministic wireless channel simulation, ray-tracing based communication modeling, and predictive analytics framework.	Live mobility trajectory data, Received Signal Strength Indicator (RSSI), Line-of-Sight (LoS) conditions, network latency measurements, wireless channel parameters, and real-time environmental sensing data.
2022	Digital Twin Network: Opportunities and Challenges	Design and architecture of Network Digital Twin for communication networks	The paper introduces the concept of a Digital Twin Network which creates a virtual model of the real communication network. The digital twin can be used for monitoring network behavior, detecting faults, testing configurations and predicting network performance without disturbing the real network.	Data-driven Digital Twin Network architecture combined with Machine Learning models. The system collects network telemetry data and uses ML-based network models to estimate performance metrics and simulate different network conditions. The DTN model acts as a virtual test environment where operators can analyze failures and evaluate configurations.	Network topology information, traffic load, routing configuration, link utilization, delay, packet drops, bandwidth usage and network performance metrics collected from real networks or simulation tools.
2024	Digital Twin Enabled Cellular Network Framework	Digital twin modeling and performance evaluation of cellular communication networks	This paper proposes a digital twin framework for cellular networks that replicates the behavior of real 4G and 5G networks using measurement data. The digital twin can analyze network performance and help improve network capacity and coverage.	Measurement-based Digital Twin modeling using real network measurements and simulation models. The framework combines real-world measurement data with digital twin simulation to evaluate network performance under different configurations such as spectrum sharing and dual connectivity.	Signal-to-interference ratio, spectral efficiency, user distribution, traffic demand, network capacity, signal strength, throughput and geographical network data.
2024	AI-Based Digital Twin Network Optimization	Improving network performance using Artificial Intelligence and Digital Twin models	The paper presents a digital twin framework integrated with Artificial Intelligence techniques to improve network performance. AI models analyze network behavior and suggest optimization strategies for better network efficiency.	Machine Learning based Digital Twin simulation where AI algorithms analyze network traffic patterns and performance metrics. The digital twin simulates different network conditions and the AI model identifies optimal network configurations for improved performance.	Traffic patterns, bandwidth utilization, network load, delay values, throughput measurements and performance indicators obtained from the digital twin simulation.
2024	Digital Twin with Multi-Agent Reinforcement Learning	Congestion reduction and optimization using Digital Twin and Reinforcement Learning	This paper integrates Digital Twin technology with Multi-Agent Reinforcement Learning to optimize system performance and reduce congestion. The digital twin simulates real-world behavior and reinforcement learning agents learn	Digital Twin simulation integrated with Multi-Agent Reinforcement Learning. Multiple agents observe system states and learn optimal actions using reward-based learning. The agents dynamically adjust system parameters to reduce congestion and improve system efficiency.	Traffic flow rates, congestion levels, vehicle counts, signal timing data, system states and performance indicators collected from simulation models.

			optimal control actions.		
2024	Digital Twin Predictive Framework	Predicting system performance using Digital Twin models and historical data	This paper develops a predictive Digital Twin system that uses historical and real-time data to predict system behavior. The model forecasts performance problems and helps in improving system reliability.	Predictive analytics based Digital Twin model using historical data analysis and statistical prediction models. The digital twin continuously updates system states and predicts future performance conditions.	Historical system data, real-time sensor data, system performance values, load measurements and environmental conditions.

Research Gap

A systematic analysis of the existing literature reveals several critical research gaps in the domain of Network Digital Twins for real-time performance diagnosis and congestion mitigation:

1. Absence of End-to-End Integrated Frameworks: While numerous studies address either digital twin modeling or congestion control independently, there is a significant lack of end-to-end integrated frameworks that combine real-time telemetry collection, digital twin synchronization, ML-based congestion prediction, and automated mitigation into a unified operational system. Most existing works focus on isolated components without providing a complete, deployable solution.
2. Insufficient Use of Historical Traffic Datasets for ML Model Training: While machine learning is increasingly applied within digital twin frameworks, limited work has systematically utilized well-established historical network traffic datasets such as the GEANT or Abilene datasets for training congestion prediction models. The absence of standardized dataset-driven ML training leads to models that may not generalize well across different network conditions and traffic patterns.
3. Absence of Mininet-Based Digital Twin Research:There is limited research specifically using Mininet as the emulation platform for constructing Network Digital Twins. Mininet provides a lightweight, programmable, and SDN-compatible network emulation environment, yet its potential for realistic digital twin implementation remains largely unexplored. Combining Mininet with real-time ML-based monitoring and SDN-driven mitigation represents an underexplored area that this project addresses directly.

Proposed Methodology

1. The proposed system will implement a Network Digital Twin using Mininet to emulate a communication network topology and replicate its behavior in a controlled environment. Network traffic will be generated using iperf to simulate realistic traffic conditions, while telemetry data such as throughput, bandwidth utilization, latency, and packet loss will be collected continuously from the emulated network.
2. Historical network traffic data from publicly available datasets such as the GEANT dataset will be analyzed to understand traffic patterns and congestion characteristics.
3. The Digital Twin environment will be used to simulate various traffic scenarios and observe their impact on network performance. Real-time telemetry collected from the emulated network will be used to diagnose and detect congestion conditions based on predefined performance thresholds.
4. Upon detecting or predicting congestion, automated mitigation techniques such as bandwidth control, traffic rerouting, or load redistribution will be applied within the Digital Twin using programmable network control mechanisms. Network performance will be continuously monitored before and after mitigation to evaluate improvements.

Possible Outcome

This integrated framework combines Digital Twin simulation, real-time telemetry using iperf, machine learning-based congestion prediction, and automated mitigation to enable intelligent network performance management.